



Motor dance creativity shows limited associations with other creative behaviours. Exploring domain-specificity in middle-aged adults

Carlota Torrents^{*}, Cristina Calvo-Estelrich

Institut Nacional d'Educació Física de Catalunya (INEFC), University of Lleida, Lleida, Spain

ARTICLE INFO

Keywords:

Embodied creativity
Creative self-efficacy
Dance improvisation
Domain specificity
Adult population
Movement variability

ABSTRACT

The extent to which motor creativity relates to overall creativity still remains unclear, especially in adults without prior experience in dance. This study explored the associations between motor dance creativity, figural creativity, and self-reported measures of creative dispositions in 71 middle-aged adults ($\text{age}_{\text{mean}} = 52.8 \pm 5.4$; 65 females and 16 males). Participants completed validated questionnaires on creative self-efficacy, emotional creativity, ideational behaviour, and lifetime creative actions, alongside two performance-based tests: a motor improvisation dance task and the Test of Creative Thinking – Drawing Production. Correlation and regression analyses were used to identify potential relationships among these measures, complemented by an exploratory factor analysis to examine their latent structure. Results revealed that motor dance creativity was significantly associated with figural creativity and creative self-efficacy, but not with other self-reports. Regression models showed that creative self-efficacy was the only significant predictor of motor dance creativity, while lifetime creative actions best predicted figural creativity. The factor analysis supported a two-factor structure, differentiating performance-based measures from self-report measures, with limited overlap. These findings offer exploratory evidence that motor dance creativity is only modestly connected to other creative behaviours, supporting its domain-specific and embodied nature. The study adds to the debate on the generality versus specificity of creativity and underscores the importance of embodied, context-dependent approaches for understanding creative behaviour.

1. Introduction

Sports sciences research has recently focused on motor creativity, the ability to generate novel and functional movement solutions, as both a potential factor in skilled performance and a trait that may enhance individual well-being (Bek et al., 2022; Memmert, 2011). However, little is known about whether the development of motor creativity extends to creativity in other areas (Richard et al., 2024), and even less about the reverse—that is, whether possessing a creative personality leads to creative motor behaviours.

The long-standing debate on whether creativity is domain-general or domain-specific remains unresolved, although recent evidence increasingly supports the latter view. Expertise significantly enhances creativity within a specific domain, while certain personality traits—such as openness to experience—also seem to promote creative expression, especially when individuals are

^{*} Corresponding author at: INEFC, Pda de la Caparrella sn, 25192 Lleida, Spain.

E-mail address: ctorrentsm@gencat.cat (C. Torrents).

intrinsically motivated (Amabile, 1996; Kaspi-Baruch, 2019). Creativity is additionally influenced by contextual factors, such as a person's current mood or enriched environments (Furley & Memmert, 2018; Memmert, 2015; Santos et al., 2016). Research has demonstrated low or modest intercorrelations among creativity tests across different domains (Baer, 2010), with stronger associations when domains are closely related (Huang et al., 2017). Consequently, creativity in one domain does not necessarily predict creativity in another (Baer, 2012); transfer is more likely when domains share overlapping features. This debate parallels principles from motor learning and coordination dynamics, where transfer is facilitated when previously acquired behaviours support the new task, but hindered when they are very different (Kelso, 1995). Interestingly, physical activity itself has been shown to foster cognitive and creative functioning in everyday contexts, suggesting potential cross-domain influences (Bollimbala & James, 2024; de Vries et al., 2022).

In motor creativity, similar to overall creativity, the emergence of creative behaviour is likely influenced by prior experiences, the similarity of the new task to previous motor learning, and personality traits that typically foster creativity, such as openness to experience, self-confidence, and motivation (Amabile, 1983; Fink & Woschnjak, 2011; Zahno, 2024). Importantly, there is a distinction between being open when judging others' behaviour and expressing that openness through one's own actions — particularly in expressive domains such as dance, which may be constrained by social prejudices that inhibit performance (Romero-Martín & Caballero-Julia, 2022). This is especially relevant when studying adult populations without prior dance experience, as they may feel more inhibited than younger populations due to emotional and social factors that further shape creative expression. Positive affective climates and supportive social interactions can facilitate exploration and flexibility, whereas negative emotional contexts may inhibit creative behaviour and lead to different outcomes in the same person (Glăveanu, 2020). Several neurobiological accounts highlight that midlife is a period during which mechanisms relevant to creative performance undergo measurable changes. Dopaminergic, serotonergic, and noradrenergic systems show progressive reductions from the fourth decade onwards (Bäckman et al., 2006), and these neuromodulatory changes have been proposed to influence processes central to divergent thinking, such as motivation, cognitive flexibility, and novelty seeking. Recent lifespan studies also indicate that divergent thinking abilities and the neural connectivity supporting them differ across adulthood, with age-related shifts between the coupling of default mode and executive control networks (Kruse et al., 2024). These patterns suggest that adults in midlife may rely on partly distinct neurocognitive resources when engaging in creative tasks. At the same time, executive functions, including cognitive flexibility, inhibition, and working memory, remain partially trainable in adulthood (Diamond, 2013), reinforcing the relevance of studying creativity in this age group, where neurobiological change coexists with preserved capacity for cognitive adaptation. Creativity is regarded as a multidimensional construct that includes cognitive, emotional, and motor processes, which interact in dynamic and complex ways with task constraints and the environment—covering both the immediate task setting and the broader sociocultural context (Glăveanu, 2020; Torrents, Balagué, Hristovski, et al., 2021a). It is frequently assessed through self-report tools (Furnham et al., 2008; Hughes et al., 2013), which measure engagement in creative activities; self-efficacy questionnaires that evaluate individuals' beliefs in their capacity to generate creative ideas or solutions (Puentes-Díaz & Cavazos-Arroyo, 2017; Tao et al., 2025); or performance-based tests that evaluate the production of ideas or solutions, with results that do not always align with self-reports (Hass et al., 2019). Motor creativity has been examined through divergent movement production tasks (Orth et al., 2017, 2019, 2024; Perez-Calzado et al., 2024; Torrents et al., 2010, 2016), through observational ratings in team sport contexts (Memmert, 2014), or via expert-based evaluations (Pérez-Calzado et al., 2026). Although some research has investigated the relationship between different types of creativity tests and motor creativity in children or young students (Büning & Lausberg, 2023; Ghanamah, 2024; Khorkova et al., 2024; Richard et al., 2018, 2021; Scibinetti et al., 2011), the connection between motor creativity in free dance improvisation and other forms of creativity in middle-aged adults without prior experience has not yet been systematically examined. Studying this enhances the possibility of using dance and improvisation for motor and cognitive benefits in this population. Improvisation tasks, often used to assess motor creativity, are particularly interesting because they require participants to manage multiple elements in real time without the opportunity to reflect in advance (Horváth et al., 2025; Richard et al., 2021; Torrents et al., 2018). Such tasks are inherently enacted and embodied, shaped by the online interaction of perception and action, and they do not require prior experience in the domain. Nevertheless, inexperienced participants may feel uncomfortable or embarrassed, suggesting that the personality traits, particularly emotional creativity, may influence outcomes.

Given the limited evidence connecting embodied creative behaviour with other creativity domains in adults, the present study adopted an exploratory, hypothesis-generating approach. Rather than testing predefined causal relationships, we aimed to examine potential associations between motor and non-motor creativity measures, identifying emerging patterns that could inform future hypothesis-driven research. For these reasons, the present study investigated the relationships between motor dance creativity and general creativity in middle-aged adults (40 to 60 years old) with no prior experience in dance or expressive movement practices. Motor creativity was assessed using both a performance-based test and a self-perceived motor creativity questionnaire, included because perceived capability is known to influence engagement and exploratory behaviour, which are central to improvisation-based motor tasks. In addition to assessing motor creativity, we included a set of complementary measures to capture different components of creative functioning: Figural creativity (TCT-DP), a well-established performance-based measure of divergent thinking that allows comparison between enacted creative movement and a non-whole-body representational task; Ideational behaviour, an indicator of everyday generative tendencies; Emotional creativity, a measure of affective flexibility that may support expressive movement; and lifetime creative actions, as a proxy for accumulated exposure to creative contexts.

Based on previous research, we hypothesised that motor creativity would show partial correlations with other creativity measures, with stronger associations expected for performance-based tests than for self-reported measures. We further hypothesised that motor creativity would partially explain variance in the other measures, while not being entirely explained by them, reflecting the specificity of creative abilities across domains.

2. Method

2.1. Participants

Participants were recruited through community advertisements and social media posts seeking individuals aged 40 to 60 years old interested in participating in a motor dance creativity study at the National Institute of Physical Education of Catalonia (INEFC) and Club INEF in the Lleida region. Individuals who expressed interest completed a brief screening to ensure that they met the inclusion criteria and had no motor or health conditions that would interfere with the task. Seventy-one adults participated voluntarily in this study ($\text{age}_{\text{mean}} = 52.8 \pm 5.4$; 65 females and 16 males). The sample reflected the natural gender distribution of individuals who voluntarily engage in dance activities in the region, and therefore represents an ecologically valid composition for this population. Eligibility criteria required participants to be between 40 and 60 years of age, to have no prior dance experience in the past five years (to reduce task-specific expertise), and to be fluent in Spanish. The study excluded individuals with recent psychiatric or neurological conditions, substance use, or psychotropic medication use during the last 5 years. Importantly, performance-based assessments such as the DICAT and TCT-DP require independent coding by trained raters, which further constrains feasible sample sizes. With 71 participants, the study provided an acceptable ratio of participants to predictors for multiple regression analyses, which exceeded the rule of thumb of 10–12 cases per predictor (Green, 1991). Additionally, it exceeded the recommended subject-to-variable ratio of at least 5:1 for exploratory factor analysis (Tabachnick & Fidell, 2019).

All participants provided written informed consent in accordance with the ethical standards outlined in the Declaration of Helsinki. The study was approved by the Ethics Committee of the Catalan Sports Council (004/CEICGC/2022).

2.2. Procedure

All participants were asked to fill out a demographic form, which asked for date of birth, sex, physical activity habits and previous experience in dance activities, as well as a series of scales assessing creative self-efficacy in movement, emotional creativity, ideation behaviours and creative activities. They were also asked to perform creativity tests of drawing production and of motor behaviour.

2.2.1. Creative self-efficacy in movement

Richard et al. (2024) developed a scale to assess perceived creative efficacy in Creative Dance, building on the theoretical framework of Beghetto and Karwowski (2017) and following Bandura's methodological guidelines for constructing self-efficacy instruments (Bandura, 2006). The initial instruction given to the participants was to "rate how confident you are that you can effectively...". Ten items were created around participants' abilities to move in original ways, express their thoughts and emotions through movement, use the environment to move creatively, and develop a sequence of movements (Richard et al., 2024). Participants rated the strength of their efficacy beliefs on a 100-point scale (0- no confidence; 100- total confidence).

2.2.2. Emotional Creativity Inventory (ECI-S)

The Spanish short version validated by Soroa et al. (2015) was used to measure emotional creativity. The ECI-S consists of a 17-item self-report questionnaire that provides information about emotional preparedness (i.e., understanding the causes/effects of emotions), novelty (i.e., the ability to experience unusual emotions), and effectiveness/authenticity (i.e., expressing emotions genuinely and honestly). Participants were asked to assess their ability to experience and express emotions on a 6-point Likert scale ranging from 1 (strongly disagree) to 6 (strongly agree). The Spanish version showed adequate internal consistency and temporal stability and confirmed the three-factor structure of the original scale.

2.2.3. Runco Ideation Behaviour Scale (RIBS)

The frequency of everyday ideational behaviours was measured using the Spanish adaptation López-Fernández et al. (2019) of the original scale by Runco et al. (2014). RIBS consists of 23 items, of which participants must indicate the frequency of occurrence of each ideational behaviour on a scale ranging from 0 (never) to 4 (very often). The Spanish version confirmed the 2-factor model and showed good discrimination and satisfactory reliability.

2.2.4. Creative Actions Scale (CAS)

The evaluation of participants' lifetime involvement in creative activities was based on the original scale developed by Elisondo (2020). The CAS consists of independent scales that assess the frequency of engagement in everyday creative actions (from never/almost never to most of the times I had the opportunity) across seven domains: Literature, Plastic Arts and Crafts, Science and Technology, Scenic Arts, Music, Social Participation, and Everyday Creativity.

2.2.5. Test of Creative Thinking – Drawing Production (TCT-DP)

Figural creativity was assessed using the test developed by Jellen and Urban (1989), which asks participants to complete drawings based on predefined fragments. The key instruction emphasised free expression without judgment: "You can draw whatever you want; whatever you put on the paper will be correct. Raters scored the productions according to the 14 criteria established by Urban (2005). For our research, two trained, independent raters—blinded to group and time assignment—carried out the assessments and reached substantial interrater reliability, with a weighted Cohen's kappa ($K_w = 0.81$), which accounts for the degree of disagreement between raters.

2.2.6. Motor creativity

Free movement creativity was assessed through a six-minute individual improvisation task in a standardised 12 × 8 m space with background music, previously used with dancers and non-dancers (Aragónés et al., 2020; Pérez-Calzado et al., 2024). In this task, participants were instructed to “move freely and explore different movement possibilities”. The Dance and Improvisation Creativity Assessment Tool (DICAT- Pérez-Calzado et al., 2026) was employed to evaluate five dimensions: quantity, referred to the number of discrete motor actions produced (e.g., a turn, a step sequence, a level change); variety, referred to the diversity of these actions, considering both the use of different categories (e.g., locomotor actions, turns, balances, jumps or changes of level), and quantitative contrasts within the same category (e.g. different body parts, rhythm or movement qualities); originality, that reflected the occurrence of unusual, unexpected or idiosyncratic actions relative to what is typically observed in novice adult improvisations; elaboration, that assessed the degree of complexity of the movement proposals, including the number of body segments involved, the execution of difficult movements or the expressive modulation of the movement; and aesthetic appeal, referred to the artistic coherence and affective impact of the improvisation considering aspects such as fluency, expressiveness and the overall impression created in the observer.

To ensure scoring consistency, raters were provided with written definitions, behavioral anchors and example interpretations for each dimension. Before formal scoring, both raters completed a calibration session using sample improvisations to align their criteria. During scoring, raters independently viewed the complete six-minute improvisation and assigned a score from 1 (“very little”) to 5 (“a lot”) on each of the five dimensions, achieving a near-perfect interrater reliability score ($Kw = 0.85$).

2.3. Data analysis

All analyses were conducted using Jamovi (version 2.6.26.0). Descriptive statistics were calculated for all variables. Bivariate associations were examined using Spearman's ρ , due to non-normality in some measures. Subsequently, two multiple linear regression models were estimated to identify predictors of (a) figural creativity and (b) motor creativity. Predictors entered simultaneously included the remaining performance-based creativity measure, creative self-efficacy, emotional creativity (ECI), ideational behaviour (RIBS), and the creative actions scale (CAS). Finally, to examine the latent structure of the creativity measures, an exploratory factor analysis (principal component extraction with oblimin rotation) was performed. Sampling adequacy was assessed using the Kaiser–Meyer–Olkin (KMO) index and Bartlett's test of sphericity. The number of components to retain was based on eigenvalues exceeding 1 and inspection of the scree plot. Statistical significance was set at $p < .05$.

All analyses were conceived as exploratory, aimed at detecting potential trends and relationships rather than confirming pre-established hypotheses. Accordingly, results were interpreted as indicative of possible mechanisms linking embodied and non-embodied creativity domains.

3. Results

3.1. Correlation analysis

Table 1 displays the Spearman correlations among creativity measures. Motor creativity was significantly associated with figural creativity ($\rho = 0.319$, $p = .007$) and creative self-efficacy ($\rho = 0.302$, $p = .010$). Figural creativity showed positive correlations with self-efficacy ($\rho = 0.293$, $p = .013$) and with lifetime creative actions (CAS; $\rho = 0.358$, $p = .002$). Exploratory analyses using the five motor creativity dimensions (quantity, variety, originality, elaboration and aesthetic interest) showed the same pattern of associations observed for the composite motor creativity score, with all dimensions being highly intercorrelated and exhibiting similar relationships with figural creativity and CAS. These results are presented in Supplementary Table S1. Strong associations were also observed among the self-report scales, particularly between RIBS and CAS ($\rho = 0.483$, $p < .001$) and between RIBS and ECI ($\rho = 0.438$, $p < .001$). Achieved power ($1 - \beta$) for all significant correlations ranged from 0.70 to 0.93, indicating adequate sensitivity to detect associations of moderate size with the current sample. Overall, performance-based measures were positively related to each other, while self-reports formed a distinct cluster, with limited overlap between the two types of measures.

Scatterplots for the main bivariate associations are provided in Supplementary Figs. S1–S3.

Table 1
Spearman correlations between creativity measures ($N = 71$).

Variable	Figural creativity	Motor creativity	Self-efficacy	ECI	RIBS	CAS
Figural creativity	–					
Motor creativity	0.319**	–				
Self-efficacy	0.293*	0.302*	–			
ECI	0.156	0.049	0.312**	–		
RIBS	0.224	0.073	0.099	0.438***	–	
CAS	0.358**	0.185	0.113	0.330**	0.483***	–

Note. ECI = Emotional Creativity Inventory; RIBS = Runco Ideational Behaviour Scale; CAS = Creative Actions Scale.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

3.2. Regression analyses

The regression predicting figural creativity explained 22.9% of the variance ($R^2 = 0.229$; see Table 2). Among the predictors, CAS emerged as the only significant positive predictor ($B = 0.15$, $p = .035$), while self-efficacy approached significance ($p = .074$). Motor creativity and the other self-report measures were not significant predictors.

The regression predicting motor creativity explained 16.6% of the variance ($R^2 = 0.166$; see Table 3). Creative self-efficacy was the only significant predictor ($B = 0.01$, $p = .023$). None of the other predictors, including figural creativity, contributed significantly to the model.

3.3. Exploratory factor analysis

The sampling adequacy was acceptable ($KMO = 0.612$), and Bartlett's test of sphericity was significant ($\chi^2(15) = 76.1$, $p < .001$). A two-factor solution was extracted, explaining 58.9% of the variance (Table 4). Factor 1 grouped all self-report measures (self-efficacy, ECI, RIBS, CAS), while Factor 2 included the two performance-based measures (motor and figural creativity). Motor creativity loaded strongly on Factor 2 (0.802), whereas figural creativity showed cross-loadings on both factors (0.624 on performance, 0.502 on self-report). The two factors were moderately correlated ($r = 0.20$), supporting a partial but limited overlap between perceived and enacted creativity.

4. Discussion

The present exploratory study examined relationships between motor dance creativity, figural creativity, and self-reported measures, including creative self-efficacy, emotional creativity, ideational behaviour, and lifetime creative actions, in middle-aged adults without prior dance experience. The hypotheses were only partially supported. As expected, motor dance creativity showed significant associations with other creativity measures, yet correlations with self-reports were weak. Regression analyses further demonstrated that the general creativity tests measured did not predict motor dance creativity, nor did motor dance creativity predict other measures, indicating limited cross-domain transfer.

The correlation analyses revealed that motor dance creativity was significantly related to figural creativity and to creative self-efficacy, but not to emotional creativity, ideational behaviour, or lifetime creative actions. Exploratory analyses using the five DICAT dimensions showed no differential associations with the other creativity measures, consistent with the interpretation that the task behaves as a unified performance construct. Figural creativity, in contrast, was more broadly connected, showing associations not only with motor dance creativity but also with creative self-efficacy and lifetime creative actions. These results suggest that creative performance in dance improvisation depends primarily on perceived ability in that specific embodied domain, while figural creativity is more strongly linked to a history of engaging in creative practices. Neurocognitive research has also paid attention to the relationship between creative thoughts and motor activity, considering that simulations within the motor system support higher order cognition, creativity included (Matheson & Kenett, 2020). Although this framework is mainly focused on the use of tools, the hypothesis that cognitive processes of generating creative thoughts are embedded in motor processes raises interesting questions about the motor creativity research and its relationship with other domains.

Büning and Lausberg (2023) also reported small correlations between motor and figural creativity (the highest being $\rho = 0.225$) in university students, somewhat lower than those observed here ($\rho = 0.319$). While both effect sizes are modest, this variability likely reflects differences in the tasks employed and the characteristics of the populations studied. Richard et al. (2021) also found weak associations between divergent thinking (verbal and figural) and motor creativity, reinforcing the idea that transfer is limited unless domains share overlapping mechanisms.

The regression analyses clarified the unique contributions of different predictors. Creative self-efficacy emerged as the only significant predictor of motor dance creativity, highlighting the role of confidence in one's creative abilities for generating novel motor solutions. In contrast, figural creativity was best predicted by lifetime creative actions, with self-efficacy approaching significance. Interestingly, neither regression analysis indicated cross-domain prediction: motor dance creativity did not predict figural creativity, nor did figural creativity predict motor dance creativity. It is also worth noting that the self-efficacy questionnaire used in this study specifically targeted the type of creativity assessed in the motor task, which may account for its predictive value. Nonetheless, the

Table 2
Multiple regression predicting figural creativity ($N = 71$).

Predictor	B	SE	t	p
Constant	7.28	5.70	1.28	0.206
Motor creativity	0.29	0.18	1.58	0.118
Self-efficacy	0.01	0.00	1.81	0.074
ECI	-0.06	0.09	-0.70	0.486
RIBS	0.05	0.08	0.67	0.507
CAS	0.15	0.07	2.16	0.035*

Model fit: $R = 0.479$, $R^2 = 0.229$.

Note. ECI = Emotional Creativity Inventory; RIBS = Runco Ideational Behaviour Scale; CAS = Creative Actions Scale.

* $p < .05$.

Table 3
Multiple regression predicting motor creativity (N = 71).

Predictor	B	SE	t	p
Constant	5.97	3.77	1.59	0.118
Figural creativity	0.13	0.08	1.58	0.118
Self-efficacy	0.01	0.00	2.33	0.023*
ECI	−0.05	0.06	−0.87	0.389
RIBS	0.01	0.05	0.11	0.910
CAS	0.03	0.05	0.64	0.527

Model fit: $R = 0.408$, $R^2 = 0.166$.

Note. ECI = Emotional Creativity Inventory; RIBS = Runco Ideational Behaviour Scale; CAS = Creative Actions Scale.

* $p < .05$.

Table 4
Exploratory factor analysis with oblimin rotation (N = 71).

Variable	Factor 1 (self-report)	Factor 2 (performance)	Uniqueness
Figural creativity	0.502	0.624	0.38
Motor creativity	0.381	0.802	0.20
Self-efficacy	0.728	–	0.46
ECI	0.706	–	0.49
RIBS	0.886	–	0.24
CAS	0.747	–	0.40

Model statistics: KMO = 0.612; Bartlett's $\chi^2(15) = 76.1$, $p < .001$.

Two-factor solution explaining 58.9% of the variance. Factor correlation = 0.20.

Note. ECI = Emotional Creativity Inventory; RIBS = Runco Ideational Behaviour Scale; CAS = Creative Actions Scale.

moderate correlation between self-efficacy and motor dance creativity indicates substantial individual variability. This suggests that self-beliefs and actual creative performance are not always aligned, as some participants displayed strong motor dance creativity despite low confidence, whereas others reported high confidence but performed less creatively. Similar results were obtained by [Pretz and McCollum \(2014\)](#), who analysed the correlation between measures of divergent thinking and specific creativity tests with creative self-efficacy in undergraduate students. Results showed that performance on creativity tasks was correlated with self-perceived creativity on specific tasks, but was unrelated to global assessments of creative self-efficacy. Nevertheless, the correlations ranged from $\rho = 0.287$ to $\rho = 0.565$, with most of them being moderate.

The factor analysis supported the existence of two distinct dimensions of creativity: self-report-based versus performance-based. While the two factors were moderately correlated, the overlap was limited, underscoring that enacted creative performance is not fully captured by self-reported tendencies or beliefs. The cross-loading of figural creativity onto both factors further suggests that it occupies an intermediate position, partly reflecting self-perceptions while still clustering with performance. Notably, motor dance creativity emerged as the most distinct measure, loading strongly on the performance factor and minimally on self-report measures, reinforcing its specificity. These findings support the theory that creative motor patterns emerge through perception-action coupling within environmental constraints ([Hristovski et al., 2011](#); [Orth et al., 2017](#); [Torrents, Balagué, Ric, & Hristovski, 2021b](#)). When the creative process requires motor activity—and particularly when motor activity itself constitutes the creative product, as is the case in all motor creativity tests—creativity emerges from the interaction between the individual and the environment. The figural creativity test also relies on such interaction, although the resulting product is external to the body (a drawing). Conversely, the motor dance creativity test evaluates the creativity of the body in motion. Both tests involve motor activity, but they differ in the degree to which the body is involved in the creative product, which may explain the moderate correlation observed. In contrast, self-report measures are abstract and decontextualised, and do not require adaptation to the environment, which may account for the larger discrepancies found.

This study therefore highlights the specificity of creativity in the sense that performance in one creative domain does not predict creativity in another. However, it also shows that when the underlying mechanisms required for creativity are related, the results tend to be associated as well. Creativity is not a unitary construct but rather a multifaceted and complex phenomenon. Motor creativity arises from the interaction between individual, environmental, and task constraints. The non-significant positive correlations obtained likely reflect how participants' personality traits influence motor creativity, but also how this influence is conditioned by prior experience (may be not specifically in free improvisation, but in motor domains) and by motor skills. This interaction is particularly evident in motor creativity, making it an excellent domain for investigating general creativity. In the present case, the participants' lack of experience in dance or improvisation further supports the generalisability of the findings, showing that even in the absence of expertise, creativity remains specific.

Finally, the use of a middle-aged sample helps to fill a gap in creativity research, which has primarily focused on children or young adults. The findings show that embodied creativity can be meaningfully assessed in populations beyond formal education or artistic training. The combination of two performance-based measures (DICAT, TCT-DP) with a battery of self-report scales provides a comprehensive view of both enacted and perceived creativity. The use of independent raters ensured a reliable assessment. Moreover, by directly comparing motor and figural creativity, the study advances the debate on domain-specificity within embodied contexts.

Some limitations should be noted. First, the sample size was modest, limiting statistical power and the generalisability of the factor analytic results; confirmatory studies with larger, more diverse samples are needed. The cross-sectional design prevents causal inferences, and the adult sample, while valuable, may not generalise to younger populations where most creativity interventions in dance are conducted. Although the exploratory factor analysis provided initial support for a distinction between self-report and performance-based measures, confirmatory approaches with larger and more diverse samples are needed to strengthen the validity of this structure.

Overall, these findings contribute to bridging creativity research and movement science by emphasizing the embodied, domain-specific nature of creative behaviour. The study offers evidence that motor creativity in middle-aged adults is only moderately linked to other forms of creativity, supporting the idea of domain-specificity. It also emphasises the importance of considering embodied, context-dependent processes when examining creativity. By combining performance-based and self-report measures, these findings help advance theoretical discussions on creativity and open new pathways for exploring how motor behaviour can inform broader insights into creative expression.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Grammarly and ChatGPT in order to improve the English writing and review the style of the paper. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Carlota Torrents: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Cristina Calvo-Estelrich:** Writing – review & editing, Visualization, Validation, Resources, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Author statement

All authors confirm that the manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have read and approved the submitted version and agree to be accountable for all aspects of the work.

Declaration of competing interest

The authors declare no conflicts of interest.

Acknowledgements

We thank Júlia Pujol and Marta Pàmies for their altruistic participation in rating the figural creativity tests, as well as Júlia Serrano and Sara Dei for their hard work and dedication in rating all the videos. We also want to thank the patience and openness of all the voluntary participants of the study.

This work was supported by a research grant awarded to Cristina Calvo-Estelrich, funded by the National Institute of Physical Education of Catalonia (INEFC) under the Government of Catalonia. The study is part of the funding program for the recruitment of early-stage researchers in the fields of physical education, physical activity, sport, and their applied sciences (PINEFC 2023), in accordance with the resolution dated 10/07/2023 and Order PRE155/23/000001.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.humov.2026.103450>.

Data availability

All anonymised data used in this study are openly available on the Open Science Framework (OSF) at doi:[10.17605/OSF.IO/3X9H7](https://doi.org/10.17605/OSF.IO/3X9H7).

References

- Amabile, T. M. (1983). The social psychology of creativity: A componential conceptualization. *Journal of Personality and Social Psychology*, 45(2), 357–376.
- Amabile, T. M. (1996). *Creativity in context: Update to the social psychology of creativity*. Routledge.

- Aragónés, M., Coterón, J., Ric, A., & Torrents, C. (2020). Influencia de la Pareja y del Espejo en la creatividad motriz de Bailarinas de Danza Contemporánea. *Revista de Psicología del Deporte*, 29(5), 107–116.
- Bäckman, L., Nyberg, L., Lindenberger, U., Li, S. C., & Farde, L. (2006). The correlative triad among aging, dopamine, and cognition: Current status and future prospects. *Neuroscience and Biobehavioral Reviews*, 30(6), 791–807. <https://doi.org/10.1016/j.neubiorev.2006.06.005>
- Baer, J. (2010). Is creativity domain specific? In J. C. Kaufman, & R. J. Sternberg (Eds.), *The Cambridge handbook of creativity* (pp. 321–341). Cambridge University Press. <https://doi.org/10.1017/cbo9780511763205.021>.
- Baer, J. (2012). Domain specificity and the limits of creativity theory. *Journal of Creative Behavior*, 46(1), 16–29. <https://doi.org/10.1002/jocb.002>
- Bandura, A. (2006). Guide for creating self-efficacy scales. In F. Pajares, & T. Urdan (Eds.), *Self-efficacy beliefs of adolescents (PB)* (pp. 307–338). Information Age Publishing.
- Beghetto, R. A., & Karwowski, M. (2017). Toward untangling creative self-beliefs. In M. Karwowski, & J. C. Kaufman (Eds.), *The creative self: Effect of beliefs, self-efficacy, mindset, and identity* (pp. 3–22). Academic Press. <https://doi.org/10.1016/B978-0-12-809790-8.00001-7>.
- Bek, J., Arakaki, A. I., Derbyshire-Fox, F., Ganapathy, G., Sullivan, M., & Poliakov, E. (2022). More than movement: Exploring motor simulation, creativity, and function in co-developed dance for Parkinson's. *Frontiers in Psychology*, 13. <https://doi.org/10.3389/fpsyg.2022.731264>
- Bollimbala, A., & James, P. S. (2024). Impact of chronic physical activity on individuals' creativity. *Psychological Research*, 88(2), 684–694. <https://doi.org/10.1007/s00426-023-01862-4>
- Bünig, C., & Lausberg, H. (2023). Divergent motor learning programs improve motor and figural creativity in university sports students. *Journal of Creative Behavior*, 57(4), 661–673. <https://doi.org/10.1002/jocb.609>
- de Vries, J. D., Bakker, A. B., & Breevaart, K. (2022). Sports lunch breaks, vigor, and creativity at work: A test of the work-home resources model. *International Journal of Sport and Exercise Psychology*, 20(6), 1594–1616. <https://doi.org/10.1080/1612197X.2021.1993960>
- Diamond, A. (2013). Executive functions. *Annual Review of Psychology*, 64, 135–168. <https://doi.org/10.1146/annurev-psych-113011-143750>
- Elisondo, R. C. (2020). Creative actions scale: A Spanish scale of creativity in different domains. *Journal of Creative Behavior*, 55(1), 2015–2227. <https://doi.org/10.1002/jocb.447>
- Fink, A., & Woschnjak, S. (2011). Creativity and personality in professional dancers. *Personality and Individual Differences*, 51(6), 754–758. <https://doi.org/10.1016/j.paid.2011.06.024>
- Furley, P., & Memmert, D. (2018). Can creative role models prime creativity in soccer players? *Psychology of Sport and Exercise*, 37, 1–9. <https://doi.org/10.1016/j.psychsport.2018.03.007>
- Furnham, A., Batey, M., Anand, K., & Manfield, J. (2008). Personality, hypomania, intelligence and creativity. *Personality and Individual Differences*, 44(5), 1060–1069. <https://doi.org/10.1016/j.paid.2007.10.035>
- Ghanamah, R. (2024). Creativity and motor skill learning among kindergarten children: Investigating predictive correlations and performance differences. *Thinking Skills and Creativity*, 54. <https://doi.org/10.1016/j.tsc.2024.101684>
- Glăveanu, V. P. (2020). The possible: A sociocultural theory. In *The possible: A sociocultural theory*. <https://doi.org/10.1093/oso/9780197520499.001.0001>
- Green, S. B. (1991). How many subjects does it take to do a regression analysis. *Multivariate Behavioral Research*, 26(3), 499–510.
- Hass, R. W., Katz-Buonincontro, J., & Reiter-Palmon, R. (2019). The creative self and creative thinking: An exploration of predictive effects using Bayes factor analyses. *Psychology of Aesthetics, Creativity, and the Arts*, 13(4), 375–387. <https://doi.org/10.1037/aca0000169>
- Horváth, M., Zuber, F., Vécsey, B. K., Varga, F., Gáspár, M., & Nagy, B. M. (2025). Move freely! – A theoretical overview of the mechanism of movement improvisation. *The Arts in Psychotherapy*, 92. <https://doi.org/10.1016/j.aip.2025.102259>
- Hristovski, R., Davids, K., Araújo, D., & Passos, P. (2011). Constraints-induced emergence of functional novelty in complex neurobiological systems: A basis for creativity in sport. *Nonlinear Dynamics, Psychology, and Life Sciences*, 15(2), 175–206.
- Huang, P. S., Peng, S. L., Chen, H. C., Tseng, L. C., & Hsu, L. C. (2017). The relative influences of domain knowledge and domain-general divergent thinking on scientific creativity and mathematical creativity. *Thinking Skills and Creativity*, 25, 1–9. <https://doi.org/10.1016/j.tsc.2017.06.001>
- Hughes, D. J., Furnham, A., & Batey, M. (2013). The structure and personality predictors of self-rated creativity. *Thinking Skills and Creativity*, 9, 76–84. <https://doi.org/10.1016/j.tsc.2012.10.001>
- Jellen, H. G., & Urban, K. K. (1989). Assessing creative potential world-wide: The first cross-cultural application of the test for creative thinking—Drawing production (TCT-DP). *Gifted Education International*, 6(2), 78–86. <https://doi.org/10.1177/026142948900600204>
- Kaspi-Baruch, O. (2019). Big five personality and creativity: The moderating effect of motivational goal orientation. *Journal of Creative Behavior*, 53(3), 325–338. <https://doi.org/10.1002/jocb.183>
- Kelso, J. A. S. (1995). *Dynamic patterns: The self-organization of brain and behavior*. MIT Press.
- Khorikova, M., Bojkowski, Ł., Korcz, A., Łopata, M., Adamczak, D., Krzysztosek, J., & Bronikowski, M. (2024). The relationship of creativity and motor creativity with physical activity and motor fitness in a gender perspective among 8–9-year-old children. *Children*, 11(12). <https://doi.org/10.3390/children11121501>
- Kruse, J. A., Martin, C. S., Hamlin, N., Slattery, E., Moriarty, E. M., Horne, L. K., ... Doucet, G. E. (2024). Changes of creative ability and underlying brain network connectivity throughout the lifespan. *Brain and Cognition*, 7(168), Article 105975. <https://doi.org/10.1016/j.bandc.2023.105975>
- López-Fernández, V., Merino-Soto, C., Maldonado Fruto, M. L., & Orozco Garavito, C. A. (2019). Analysis of the descriptive and psychometric characteristics of the internal structure of the Ribs in Spanish. *Creativity Research Journal*, 31(2), 229–235. <https://doi.org/10.1080/10400419.2019.1577123>
- Matheson, H. E., & Kenett, Y. N. (2020). The role of the motor system in generating creative thoughts. *NeuroImage*, 213. <https://doi.org/10.1016/j.neuroimage.2020.116697>
- Memmert, D. (2011). Sports and creativity. In E.-C. M. A. Runco, & S. R. Pritzker (Eds.), *Encyclopedia of creativity* (2nd ed., pp. 373–378). Academic Press. <https://doi.org/10.1016/B978-0-12-375038-9.00207-7>.
- Memmert, D. (2014). Tactical creativity in team sports. *Research in Physical Education, Sport and Health*, 3, 13–18.
- Memmert, D. (2015). Teaching tactical creativity in sport: Research and practice. In *Teaching tactical creativity in sport: Research and practice*. Taylor and Francis Inc.
- Orth, D., Van der Kamp, J., Memmert, D., & Savelsbergh, G. J. P. (2017). Creative motor actions as emerging from movement variability. *Frontiers in Psychology*, 8 (October), 1–8. <https://doi.org/10.3389/fpsyg.2017.01903>
- Orth, D., McDonic, L., Ashbrook, C., & van der Kamp, J. (2019). Efficient search under constraints and not working memory resources supports creative action emergence in a convergent motor task. *Human Movement Science*, 67, Article 102505. <https://doi.org/10.1016/j.humov.2019.102505>
- Orth, D., van der Kamp, J., Memmert, D., & Savelsbergh, G. (2024). Skill shapes functional movement variability and exploration tendencies: The effect of skill under changing environmental constraints on the emergence of creative action in a divergent kickboxing task. *International Journal of Sport and Exercise Psychology*, 22 (2), 446–466. <https://doi.org/10.1080/1612197X.2023.2280108>
- Pérez-Calzado, E., Coterón, J., Ric, Á., & Torrents, C. (2024). Analysis of the motor behaviour of people without specific dance training in improvisation tasks: Influence of mirror and partner. *Journal of Sports Sciences*, 24, 1–11. <https://doi.org/10.1080/02640414.2024.2306064>
- Pérez-Calzado, E., Calvo-Estelrich, C., Torrents, C., & Coterón, J. (2026). Validation of the DICAT instrument for assessing motor creativity in dance improvisation tasks. *Apunts, Educació Física i Esports*, 163.
- Pretz, J. E., & McCollum, V. A. (2014). Self-perceptions of creativity do not always reflect actual creative performance. *Psychology of Aesthetics, Creativity, and the Arts*, 8(2), 227–236. <https://doi.org/10.1037/a0035597>
- Puente-Díaz, R., & Cavazos-Arroyo, J. (2017). Creative self-efficacy: The influence of affective states and social persuasion as antecedents and imagination and divergent thinking as consequences. *Creativity Research Journal*, 29(3), 304–312. <https://doi.org/10.1080/10400419.2017.1360067>
- Richard, V., Lebeau, J. C., Becker, F., Boiangin, N., & Tenenbaum, G. (2018). Developing cognitive and motor creativity in children through an exercise program using nonlinear pedagogy principles. *Creativity Research Journal*, 30(4), 391–401. <https://doi.org/10.1080/10400419.2018.1530913>
- Richard, V., Ben-Zaken, S., Siekańska, M., & Tenenbaum, G. (2021). Effects of movement improvisation and aerobic dancing on motor creativity and divergent thinking. *Journal of Creative Behavior*, 55(1), 255–267. <https://doi.org/10.1002/jocb.450>

- Richard, V., Brown, D. M. Y., Garcías, S., Almarcha, M., Cairney, J., & Torrents, C. (2024). The exploration of the holistic and complex impacts of creative dance on creative potential enhancement. *Journal of Creative Behavior*, 58(3), 511–529. <https://doi.org/10.1002/jocb.673>
- Romero-Martín, M. R., & Caballero-Julia, D. (2022). Pre-service university training, body expression and self-concept. *International Journal of Environmental Research and Public Health*, 19(23). <https://doi.org/10.3390/ijerph192316218>
- Runco, M. A., Walczyk, J. J., Acar, S., Cowger, E. L., Simundson, M., & Tripp, S. (2014). The incremental validity of a short form of the ideational behavior scale and usefulness of distractor, contraindicative, and lie scales. *Journal of Creative Behavior*, 48(3), 185–197. <https://doi.org/10.1002/jocb.47>
- Santos, S. D. L., Memmert, D., Sampaio, J., & Leite, N. (2016). The spawns of creative behavior in team sports: A creativity developmental framework. *Frontiers in Psychology*. <https://doi.org/10.3389/fpsyg.2016.01282>
- Scibinetti, P., Tocci, N., & Pesce, C. (2011). Motor creativity and creative thinking in children: The diverging role of inhibition. *Creativity Research Journal*, 23(3), 262–272. <https://doi.org/10.1080/10400419.2011.595993>
- Soraa, G., Gorostiaga, A., Aritzeta, A., & Balluerka, N. (2015). A shortened Spanish version of the emotional creativity inventory (the ECI-S). *Creativity Research Journal*, 27(2), 232–239. <https://doi.org/10.1080/10400419.2015.1030313>
- Tabachnick, B. G., & Fidell, L. S. (2019). *Using multivariate statistics*. Pearson.
- Tao, R., Zhang, H., Geng, L., Li, Y., & Qiu, J. (2025). The influence of trait and state creative self-efficacy on creative behavior: An experimental study using false feedback. *Behavioral Science*, 15(1). <https://doi.org/10.3390/bs15010018>
- Torrents, C., Castañer, M., Dinușová, M., & Anguera, M. T. (2010). Discovering new ways of moving: Observational analysis of motor creativity while dancing contact improvisation and the influence of the partner. *The Journal of Creative Behavior*, 44(1), 53–69. <https://doi.org/10.1002/j.2162-6057.2010.tb01325.x>
- Torrents, C., Ric, A., Hristovski, R., Torres-Ronda, L., Vicente, E., & Sampaio, J. (2016). Emergence of exploratory, technical and tactical behavior in small-sided soccer games when manipulating the number of teammates and opponents. *PLoS One*, 11(12), Article e0168866. <https://doi.org/10.1371/journal.pone.0168866>
- Torrents, C., Ensenyat, A., Ric, A., Mateu, M., & Hristovski, R. (2018). Free play with equipment to foster exploratory behavior in preschoolers. *Nonlinear Dynamics, Psychology, and Life Sciences*, 22(4), 509–533.
- Torrents, C., Balagué, N., Hristovski, R., Almarcha, M., & Scott Kelso, J. A. (2021a). Metastable coordination dynamics of collaborative creativity in educational settings. *Sustainability*, 13(5), 1–16. <https://doi.org/10.3390/su13052696>
- Torrents, C., Balagué, N., Ric, A., & Hristovski, R. (2021b). The motor creativity paradox: Constraining to release degrees of freedom. *Psychology of Aesthetics, Creativity, and the Arts*, 15(2), 340–351. <https://doi.org/10.1037/aca0000291>
- Urban, K. K. (2005). Assessing creativity: The test for creative thinking-drawing production (TCT-DP). *International Education Journal*, 6(2), 272–280.
- Zahno, S. (2024). Creativity in sports: The crucial role of motor skills. *Journal of Creativity*, 34(3). <https://doi.org/10.1016/j.yjoc.2024.100091>